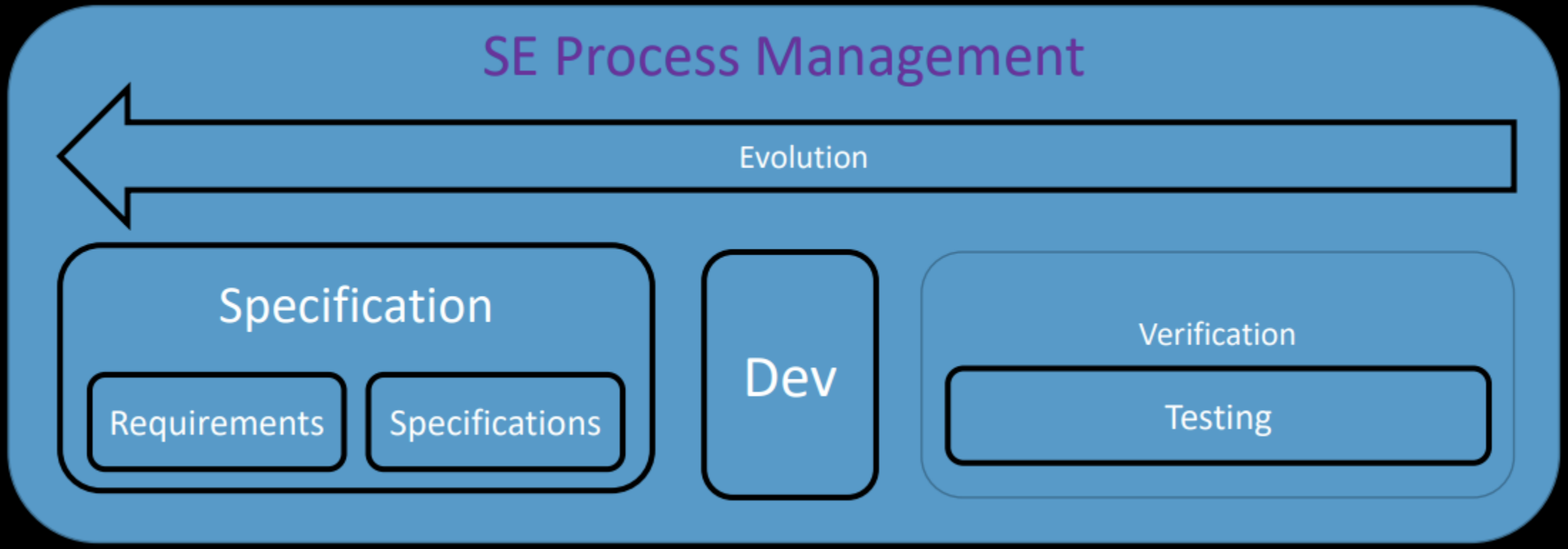
Software Engineering Management COMP4135

Seminar 7

Agile Method

Agile

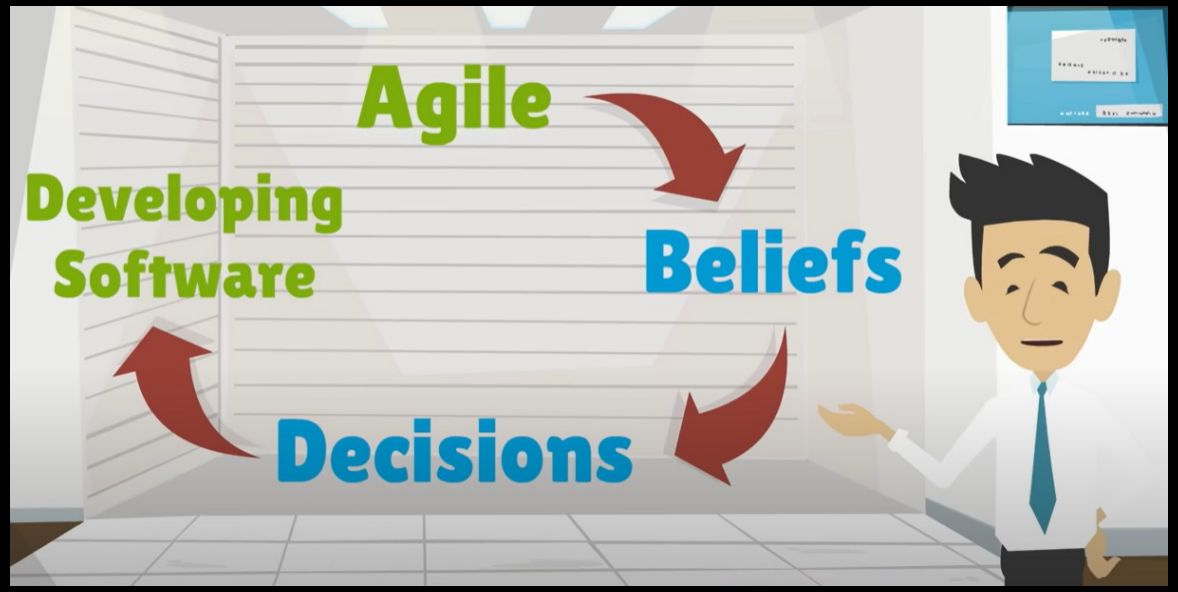

SE Process Management
Evolution
Specification
Requirements
Specifications
Dev
Verification
Testing

Agile Principles

Agile is a set of values and principles

Working software over comprehensive documentation

Responding to change over following a plan



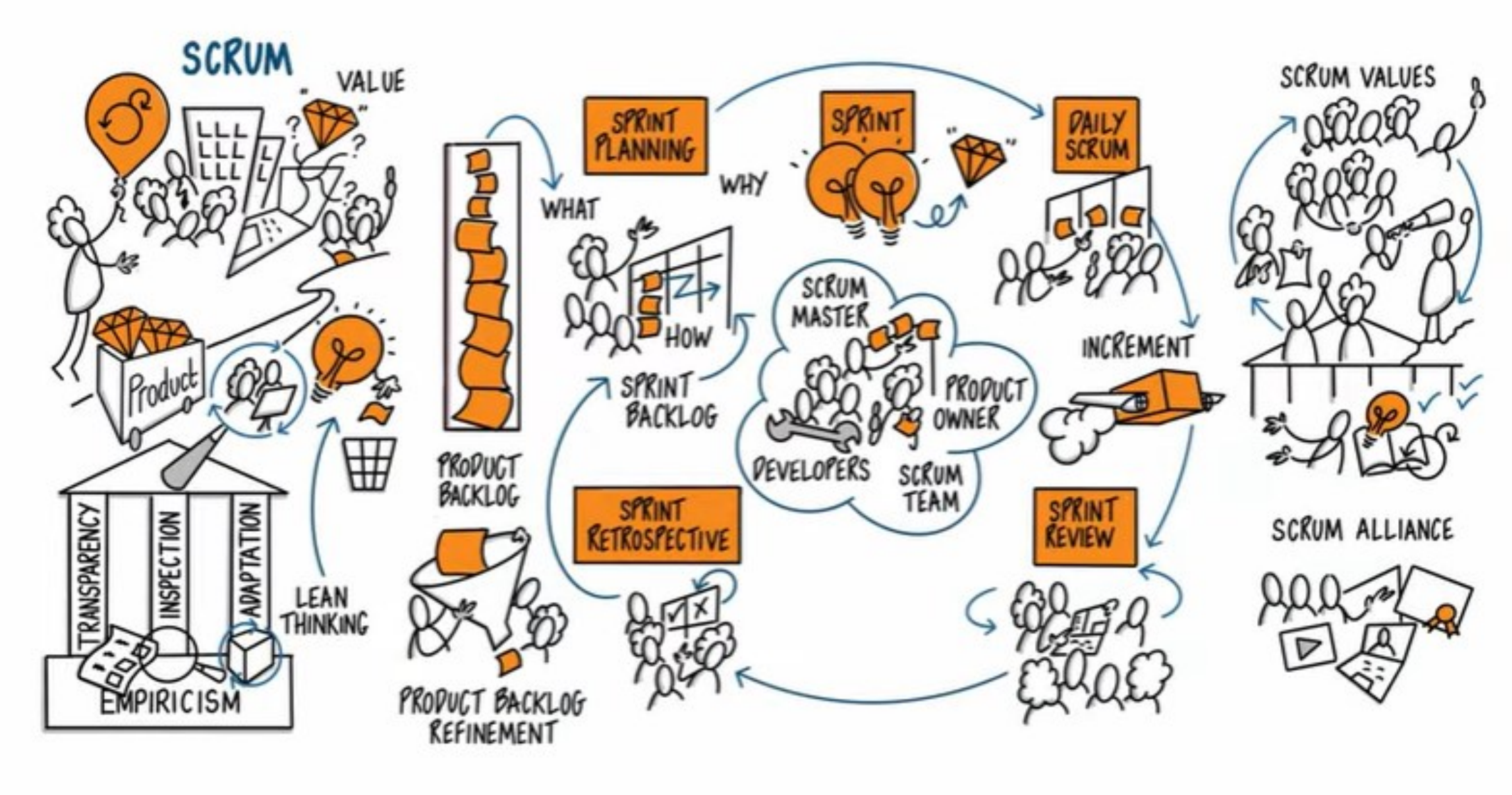
Agile Principles

Principle or practice	Description
Incremental planning	Requirements are recorded on Story Cards and the Stories to be included in a release are determined by the time available and their relative priority. The developers break these Stories into development 'Tasks'
Small releases	The minimal useful set of functionality that provides business value is developed first. Releases of the system are frequent and incrementally add functionality to the first release
Simple design	Enough design is carried out to meet the current requirements and no more.
Test-first development	An automated unit test framework is used to write tests for a new piece of functionality before that functionality itself is implemented.
Refactoring	All developers are expected to refactor the code continuously as soon as possible code improvements are found. This keeps the code simple and maintainable.
Pair programming	Developers work in pairs, checking each other's work and providing the support to always do a good job
Collective ownership	The pairs of developers work on all areas of the system, so that no islands of expertise develop and all the developers take responsibility for all of the code. Anyone can change anything.
Continuous integration	As soon as the work on a task is complete, it is integrated into the whole system. After any such integration, all the unit tests in the system must pass.
Sustainable pace	Large amounts of overtime are not considered acceptable as the net effect is often to reduce code quality and medium term productivity
On-site customer	A representative of the end-user of the system (the Customer) should be available full time for the use of the XP team. In an extreme programming process, the customer is a member of the development team and is responsible for bringing system requirements to the team for implementation

Scrum

- A lightweight framework that helps people, teams and organizations generate value through adaptive solutions for complex problems.
- Concentrate particularly on how to manage tasks within a **team-based** development environment.

Scrum


SCRUM
VALUE
Product
TRANSPARENCY
INSPECTION
ADAPTATION
EMPIRICISM
LEAN THINKING
PRODUCT BACKLOG
PRODUCT BACKLOG REFINEMENT
WHAT
SPRINT PLANNING
WHY
SPRINT
HOW
SPRINT BACKLOG
SCRUM MASTER
PRODUCT OWNER
DEVELOPERS
SCRUM TEAM
SPRINT RETROSPECTIVE
DAILY SCRUM
INCREMENT
SPRINT REVIEW
SCRUM VALUES
SCRUM ALLIANCE

Scrum Roles

- **ScrumMaster**: the keeper of the scrum process, responsible for
  - Making the process run smoothly
  - Removing obstacles that impact productivity
  - Organizing and facilitating the critical meetings
- **Product Owner**: the sole person responsible for managing the Product Backlog
  - Expressing Product Backlog items clearly
  - Ordering the Product Backlog items to best achieve goals and missions
  - Optimizing the value of the work the Team performs
  - Ensuring that the Product Backlog is visible, transparent, and clear to all, and shows what the Team will work on further
  - Ensuring that the Team understands items in the Product Backlog to the level needed
- **Dev Team:** comprises of analysts, designers, developers, tester, as relevant to the project

Scrum Artifact

- **Product Backlog:**
  - An ordered list of features that are needed as part of the end product
  - Can be updated at any time by the Product Owner
- **Sprint Backlog:**
  - Set of Product Backlog items selected for the Sprint, plus a plan for delivering the product Increment and realizing the Sprint Goal
- **Increment:**
  - Sum of all the Product Backlog items completed during a Sprint combined with the increments of all previous Sprints.

Scrum events

1. Sprint
   Timeboxed core (1-4 weeks) for delivering increments.

2. Sprint Planning
   Define goal, select tasks, plan execution (≤8 hours).

3. Daily Scrum
   15-min sync: progress, blockers, adapt plans.

4. Sprint Review
   Demo outcomes, update backlog with stakeholders.

5. Retrospective
   Reflect & improve team processes.

The Scrum Framework At a Glance

Product Backlog

(owned by PO)

The single source of work for the team, the product backlog is an emergent, ordered list of what is needed to improve the product and includes the product goal.

Refinement

Refinement is an ongoing activity used to add details, including size and order, to items in the product backlog.

Sprint Planning

To start the sprint, sprint planning establishes the sprint goal, what can be done, and how the chosen work will be completed.

Sprint Backlog

(owned by the developers)

The sprint backlog is a real-time, emergent picture of work that shows the product backlog items selected for the sprint by the developers, plus a plan for delivering the product increment and realizing the sprint goal.

*Sprint of 1 month or less

(An inspect and adapt cycle that turns ideas into value)

Each sprint should bring the product closer to the overall product goal.

Daily Scrum

The purpose of the daily scrum is for developers to inspect progress toward the sprint goal and adapt the sprint backlog as necessary, adjusting the upcoming planned work.

Sprint Review

The purpose of the sprint review is to inspect the outcome of the sprint and progress toward the product goal with key stakeholders, while determining future adaptations.

Increment

A sum of usable sprint backlog items completed by the developers in the sprint that meets the definition of done. Each increment should work with all prior increments.

Sprint Retrospective

The sprint retrospective is for the scrum team to inspect how the last sprint went with regards to individuals, interactions, processes, tools, and definition of done. The team identifies improvements to effectiveness, addressing the most impactful improvements as soon as possible.

Product Owner | **Scrum Master** | **Developers**

Scrum artifacts that help manage the work | Events that occur inside each sprint | Ongoing activity

Who should participate and for how long?

SPRINT PLANNING	DAILY SCRUM	SPRINT REVIEW	SPRINT RETROSPECTIVE
Who: The entire scrum team. **Timebox:** Maximum of 8 hours for a month-long sprint. Shorter timebox for shorter sprints.	**Who:** The developers, but if the product owner or scrum master are actively working on items in the sprint backlog, they participate as developers. **Timebox:** 15 minutes.	**Who:** The entire scrum team. Stakeholders are invited to provide feedback on the increment. **Timebox:** Maximum of 4 hours for a month-long sprint. Shorter timebox for shorter sprints.	**Who:** The entire scrum team. **Timebox:** Maximum of 3 hours meeting for a month-long sprint. Shorter timebox for shorter sprints.

Scrum Alliance®

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Agile vs Traditional - Comparison

Traditional Methods	Agile Methods
Waterfall, V-Model	Spiral, Iterative Models
Requirements Tables	User Stories, Personas
Detailed UML/Specs	Rapid Prototyping
Integration Testing Phase	TDD, Paired Coding
Formal User Testing	Frequent Client Interaction
Separate Skill Teams	Integrated skills teams

Scrum Management – Agile Simulation

Case Study: Prescribing Medication

Kate is a doctor who wishes to prescribe medication for a patient attending a clinic. The patient record is already displayed on her computer so she clicks on the medication field and can select current medication', 'new medication' or 'formulary'.
If she selects 'current medication', the system asks her to check the dose. If she wants to change the dose, she enters the dose and then confirms the prescription.
If she chooses 'new medication', the system assumes that she knows which medication to prescribe. She types the first few letters of the drug name. The system displays a list of possible drugs starting with these letters. She chooses the required medication and the system responds by asking her to check that the medication selected is correct. She enters the dose and then confirms the prescription.
If she chooses 'formulary', the system displays a search box for the approved formulary. She can then search for the drug required. She selects a drug and is asked to check that the medication is correct. She enters the dose and then confirms the prescription.
The system always checks that the dose is within the approved range. If it isn't, Kate is asked to change the dose.
After Kate has confirmed the prescription, it will be displayed for checking. She either clicks 'OK' or 'Change'. If she clicks 'OK', the prescription is recorded on the audit database. If she clicks on 'Change', she reenters the 'Prescribing medication' process.

Scrum Management – Group Discussion

Your team will develop a clinic's prescribing system with 3 workflows:

- Current meds (dose adjustment)
- New meds (search/validation)
- Formulary (approved drug list)

Discussion 1: Divide the scrum roles in your group,
How many people are needed for each role?

- Product Owner
- Scrum Master
- Dev Team

Discussion 2: Sprint Planning - Pick Your Backlog!

1. Which workflow is riskiest? Why?
2. Scoring the Backlog Items: ([High] [Mid] [Low])
    - Dose editor
    - Drug search
    - Audit log

Scrum Management – Group Discussion

Mid-Sprint Chaos (Injected Change)

"Add allergy warnings before prescribing. Adjust your Sprint plan!“
How will this impact your goal ?

Sprint Review (Demo & Feedback)

1. Show your paper prototype/wireframe.
2. PO: "Does it pass dose validation?"
3. Stakeholders: Give feedback!

Sprint 2 – XP Twist

Design the tests for the tasks

Scrum Management – Group Discussion

For the following works, match the work with the roles and scrum events:

- Prioritize the backlog: "Dose validation feature must be Sprint 1."
- Define acceptance criteria: "All prescriptions require dose-range checks before submission."
- Balance compliance vs. speed: "Delay fancy UI to implement allergy checks first.“
- Enforce timeboxes: “Limit planning to 1 hour.”
- Educate on Agile: "Explain why we timebox planning.“
- Estimate effort: Use Planning Poker for drug-search tasks
- Clarify requirements: "Doctors need drug synonyms in search." (if asked)
- Observe progress (but do not direct tasks)
- Remove blockers: *"Coordinate with IT to fix API latency.“*
- Enforce timeboxes: *"End at 15 mins—no tech debates!“*
- Update impediment board: *"Drug name ambiguity unresolved.“*
- Report progress:*"Yesterday: Built dose checker."Today: Fix search bug.“Blocker: Need drug DB docs.“*
- Demo to stakeholders: Show dose validation workflow
- Approve/reject work: *"Reject audit log without timestamps.*
- Gather feedback: *"Interview doctors about search pain points.“*
- Ensure actionable feedback: *"Turn 'slow search' into backlog items.“*
- Track action items: *"Add 'optimize queries' to next Sprint.“*
- Demo features: Live prototype of drug search.
- Explain tech debt: *"Legacy DB slows queries*
- Share stakeholder pain: "Doctors hate slow search!“
- **Advocate for quality**: *"We need more test coverage.“*
- **Facilitate discussion**: *"Why did dose tests fail?“*
- **Suggest improvements:** *"Adopt CI pipelines "Pair programming for complex logic."*
- **Shield from distractions**: *"Postpone non-urgent admin requests.“*
- **Detail user stories**: *"Add 'patient allergy pop-up’.*
- **Rank by risk**: *"Dose checker > UI polish.“*
- **Prevent scope creep**: *"Is 'fancy animations' necessary?“*
- **Clarify estimates**: *"How long for audit log setup?“*
- **Technical breakdown**: *"Optimize formulary queries."Set up audit log table (user/dose/timestamp).“*
- **Unblock progress**: *"Get DB schema docs from IT.“*
- **Execute tasks**:*"Code dose checker (morphine ≤10mg).Write allergy warning tests. "Connect UI to drug API"Ensure audit logs trigger on confirmation."*

Scrum Management – Group Discussion

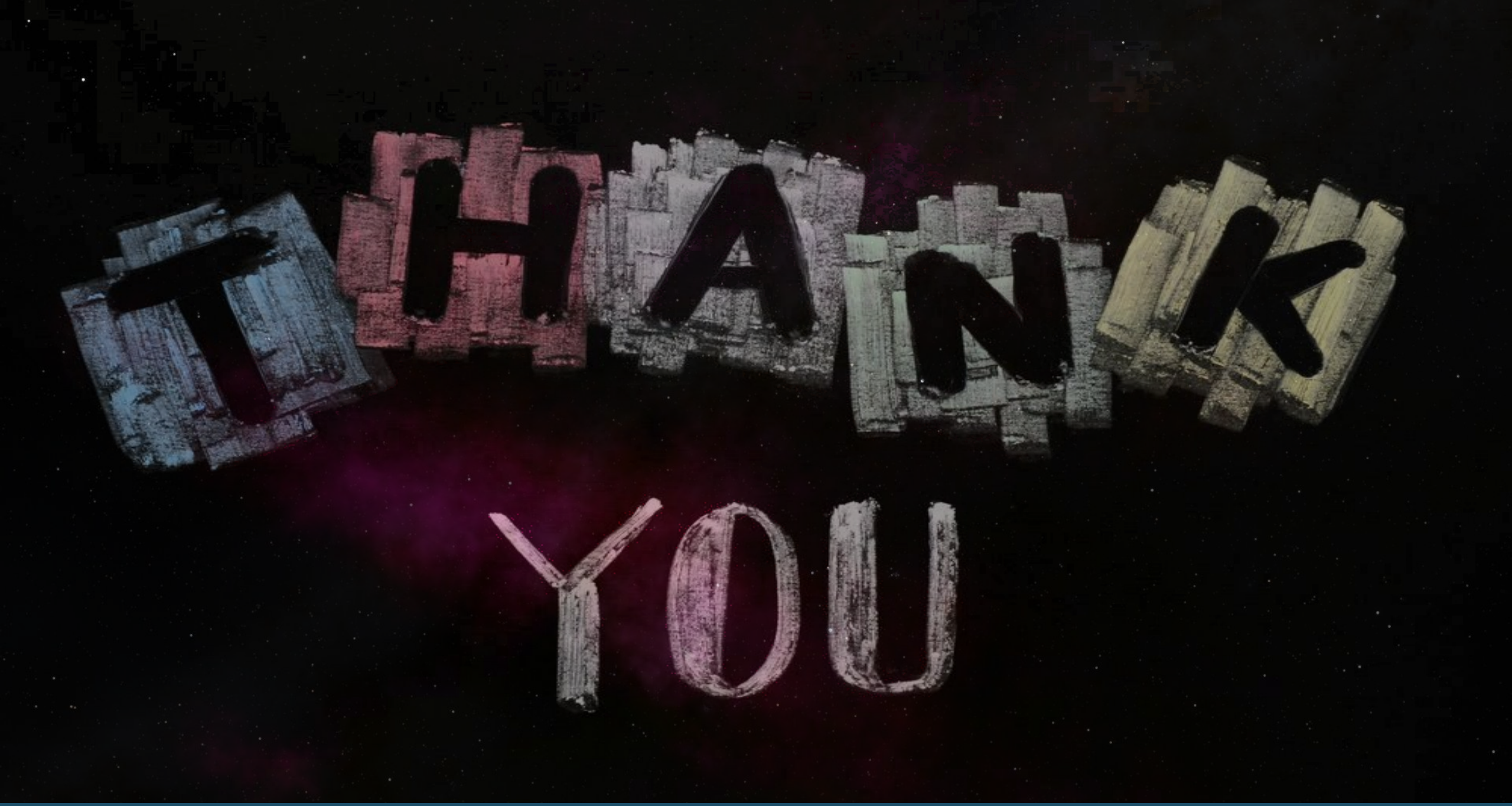
Scrum Event	Product Owner (PO) Behaviors	Scrum Master (SM) Behaviors	Development Team Behaviors
Sprint Planning			
Daily Scrum			
Sprint Review			
Sprint Retrospective			
Backlog Refinement			
Development			

Scrum Event	Product Owner (PO) Behaviors	Scrum Master (SM) Behaviors	Development Team Behaviors
Sprint Planning	3. Prioritize the backlog 5. Define acceptance criteria 12. Balance compliance vs. speed 29. Rank by risk 25. Estimate effort	7. Educate on Agile 19. Enforce timeboxes 14. Prevent scope creep 30. Clarify estimates	10. Technical breakdown 25. Estimate effort (joint with PO)
Daily Scrum	17. Clarify requirements 31. Observe progress	2. Enforce timeboxes 4. Remove blockers 8. Update impediment board 22. Unblock progress	18. Report progress
Sprint Review	6. Demo features 11. Gather feedback 15. Approve/reject work 26. Demo to stakeholders	23. Ensure actionable feedback 20. Track action items	6. Demo features 26. Demo to stakeholders 27. Explain tech debt
Sprint Retrospective	9. Share stakeholder pain 24. Advocate for quality	16. Facilitate discussion 28. Shield from distractions	13. Suggest improvements
Backlog Refinement	21. Detail user stories 29. Rank by risk	14. Prevent scope creep 30. Clarify estimates	10. Technical breakdown
Development	-	4. Remove blockers 22. Unblock progress	1. Execute tasks 10. Technical breakdown 13. Suggest improvements 27. Explain tech debt

Assessment !!! Week Report 3

Reminder:

- Submission Deadline: 17 April 2026, 17:00
- Follow the template on Moodle
- Only team leader need to submit the team report
- The team report link should also shown in your Team GitLab


THANK
YOU